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OmniFusion Technical Report

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Abstract

Last year, multimodal architectures served up a revolution in AI-based approaches and solutions, extending the capabilities of large language models (LLM). We propose an *OmniFusion* model based on a pretrained LLM and adapters for visual modality. We evaluated and compared several architecture design principles for better text and visual data coupling: MLP and transformer adapters, various CLIP ViT-based encoders (SigLIP, InternVIT, etc.), and their fusing approach, image encoding method (whole image or tiles encoding) and two 7B LLMs (the proprietary one and open-source Mistral). Experiments on 8 visual-language benchmarks show the top score for the best OmniFusion setup in terms of different VQA tasks in comparison with open-source LLaVA-like solutions: VizWiz, Pope, MM-Vet, ScienceQA, MMBench, TextVQA, VQAv2, MMMU. We also propose a variety of situations, where OmniFusion provides highly-detailed answers in different domains: housekeeping, sightseeing, culture, medicine, handwritten and scanned equations recognition, etc. Mistral-based OmniFusion model is an open-source solution with weights, training and inference scripts available at https://github.com/AIRI-Institute/OmniFusion.

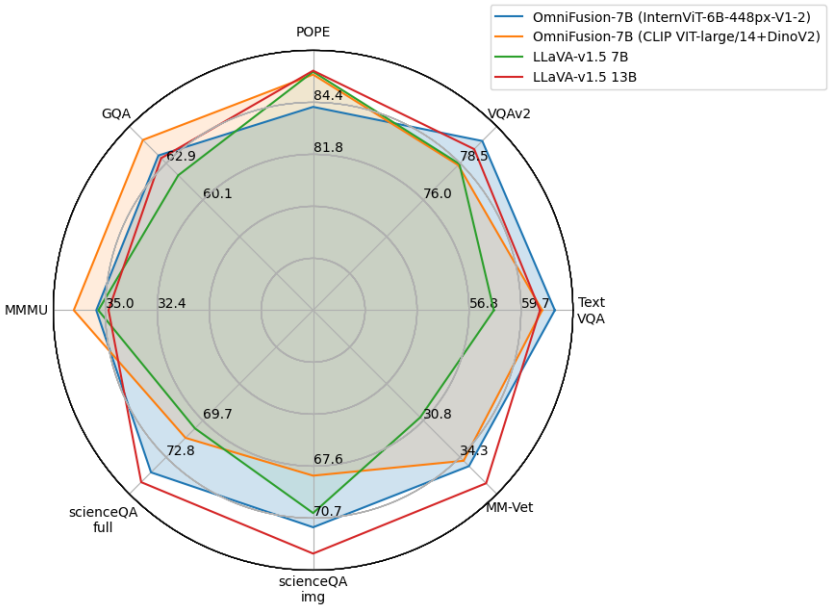

Figure 1: Comparison of OmniFusion performance on the benchmarks and generation examples.

Figure 2: OmniFusion VQA examples.

1 Introduction

In recent years, multimodal architectures emerged as a powerful paradigm for enhancing artificial intelligence (AI) systems, enabling them to process and understand multiple types of data simultaneously [1, 2, 3]. The integration of different data modalities, such as text and images, has significantly improved the capabilities of large language models (LLMs) in various tasks, ranging from visual question answering (VQA) [4] to complex decision-making processes [5, 6]. However, the challenge of effectively coupling various data types remains a significant obstacle in the development of truly integrative AI models. Furthermore, such multimodal multitask architectures are interpreted as the first steps towards the development of the artificial general intelligence (AGI), expanding the number of challenges in world cognition.

This work introduces the *OmniFusion* model, a novel multimodal architecture that leverages the strengths of pretrained LLMs and introduces specialized adapters for processing visual information. We evaluated multiple architectural designs to fuse text and visual data, such as MLP and transformer adapters. At the same time we concentrated on the comparison of various image encoders like CLIP-ViT [7] and SigLIP [8] and on visual encoders fusion techniques. The evaluated approaches gave us a broad scope of improving the visual context for better visual information extraction and retrieval.

One of the key innovations of OmniFusion is its flexible approach to image encoding, exploring both the whole image and the tiled image encoding strategies, which allows for a more nuanced understanding of visual content in relation to textual data [9]. This adaptability is critical in addressing the diverse requirements of visual-language benchmarks, where OmniFusion has demonstrated superior performance across a range of VQA tasks, outperforming existing open-source solutions.

Our extensive evaluations on eight visual-language benchmarks, including VizWiz [10], POPE [11], MM-Vet [12], ScienceQA [13], MMBench [14], TextVQA [15], VQAv2 [16], and MMMU [17], confirm the effectiveness of the OmniFusion model. The model not only achieves competitive performance with the leading approaches in various VQA tasks but also excels in providing detailed answers across multiple domains, such as housekeeping, sightseeing, culture, and medicine, showcasing its versatility and broad applicability.

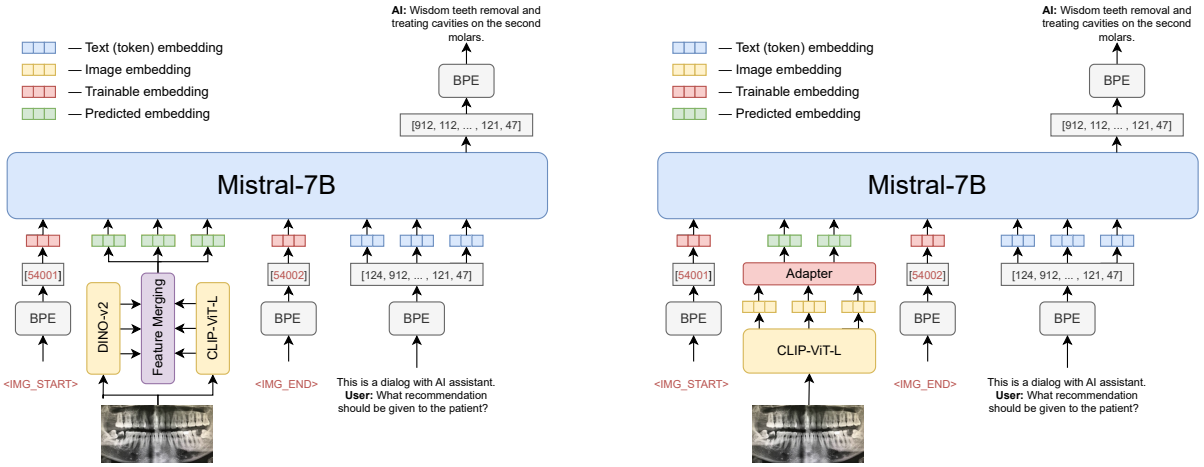

Figure 3: OmniFusion architecture with feature merging (left) and with single adapter (right): MLP or transformer layer.

2 OmniFusion

2.1 Model Architecture

The OmniFusion model integrates a pretrained LLM with special adapters for image embeddings, facilitating the fusion of the text and visual modalities. This approach has already proved its applicability in various VLM developments because the adapter approach is undemanding of computational resources in comparison with end-to-end training pipelines. Moreover, it does not need large interleaved text-image datasets to be trained, while for end-to-end approaches such data is extremely required. Overall, we have two major points that were investigated and covered further in this report: adapter technique selection and encoding visual data strategy determination.

We used special tokens with trainable embeddings to signify the beginning and the end of a token sequence from the non-text modality. This approach enhances the model's ability to distinguish between textual and visual data streams. The visual encoder's output embeddings are processed either by a transformer adapter, consisting of a single transformer layer with four heads, or through a two-layer MLP.

Our experiments showed the most effective performance with a merging strategy that combines features of two encoders: CLIP-ViT-L and DINO-v2. The merging adapter is essentially a two-layer MLP, where the first layer is independent for each visual token, and the output layer is shared across all tokens. The merging strategy implies summing up the outputs after the first layer, optimizing the integration of textual and visual information (Figure 3).

2.2 Training pipeline

When the overall architecture setup is fixed, we continue with overall training, which includes determination of the stages and the used datasets. The OmniFusion model undergoes a two-stage training process designed to harness its multimodal capabilities efficiently.

Stage 1: Pretraining. In the first stage, adapters and special tokens undergo pretraining on a vast dataset of image-text pairs. This phase aims to fine-tune the adapters for transforming visual embeddings and training special tokens to mark the transition between text and image data.

During this stage, we leverage image-text pairs tailored specifically for image captioning purposes. The image-captioning data we employ is sourced from ShareGPT4V-PT, which is supported by existing captioning datasets such as COCO and TextCaps. Additionally, we incorporate proprietary datasets containing document descriptions obtained through optical character recognition (OCR) in the pretraining set.

The instructions provided to the model during pretraining, include variations of prompts such as "Give a brief description of the image," "Describe the image in detail," and "Provide a short description of the image."

It's worth noting that we currently do not employ interleaved data during pretraining, which could potentially enhance the few-shot learning capabilities of multimodal models. However, we plan to explore this avenue in our future work.

We use the following sources of image-text captions: ShareGPT4V-PT (695K pairs), LAION-CC-SBU with BLIP captions (558K pairs) [18]. Overall, we utilize 1.2M image captions during the pretraining phase for the adapter alignment.

Stage 2: Fine-tuning. The second stage entails fine-tuning the multimodal model using a set of instructional dialogues. This process aims to enhance the model's ability to comprehend and respond to complex queries that require an integrated analysis of textual and visual information. Moreover, it is structured to facilitate multi-task fine-tuning in accordance with the natural instructions.

Recent research [19] has highlighted the potential pitfalls associated with the synthetic nature of data used for visual instruction tuning, as well as the lack of task diversity, which can lead to significant hallucinations in the output of multimodal models. To mitigate these challenges, we construct the supervised fine-tuning dataset by combining academic benchmarks with corresponding instructions and synthetic conversational data. This approach ensures a more diverse and robust training environment for the model. Table 1 provides the datasets utilized in the fine-tuning procedure.

Table 1: Distribution of the datasets for supervised fine-tuning.

Task	Dataset Source	Samples
Caption	ShareGPT4V [20]	100K
VQA	COCO [21], SAM-9K [22]	20K, 9K
WebQA	WebData [23]	1.5K
OCRQA	TextVQA [24], OCRVQA [25]	120K
Conversation	LLaVA-v1.5-665K [18]	665K
DocVQA	Proprietary data (ru)	20K
Text-only SFT	Proprietary data (ru), Alpaca (en)	10K

Training hyperparameters. During the training procedure, we utilize a standard approach with a learning rate of 1e-3 and a batch size of 256 for the pretraining phase. For the SFT phase, we adjust the learning rate to 2e-5 and the batch size to 128. We train the model using the bf16 precision, employing the AdamW optimizer with a weight decay set to 0.

The sequence length is initially set to 2048 for the base experiment, and we extend it to 4096 for experiments involving grid splitting. All experiments are conducted using 8 Nvidia A100-80Gb GPUs.

Table 2: Training Procedure Parameters

Parameter	**Pretraining Phase**	**SFT Phase**
Learning Rate	1e-3	2e-5
Batch Size	256	128
Precision	bf16	
Optimizer	AdamW	
Weight Decay	0	
Sequence Length	2048 (Base), 4096 (Grid Splitting)	
Hardware	8 Nvidia A100-80Gb GPUs	

3 Experiments

3.1 Experimental setup

Vision encoders. The influence of the image encoders is inevitable for multimodal systems. Basically, vision encoders are kept frozen via image-text alignment, so, its initial capabilities are crucial for further performance of LMMs.

In our experiments, we evaluated several encoders varying in size, training data distribution, and image resolution. Overall, the visual encoders that we tested in the experiments, are listed below:

1. CLIP ViT-bigG/14[1]. A CLIP ViT-bigG/14 model was trained with the LAION-2B English subset of LAION-5B [26] following the openclip architecture [27].

[1] https://github.com/mlfoundations/open_clip

2. CLIP ViT-L/14 [7] from openai/clip-vit-large-patch14. This is the vision encoder that was retrieved from the CLIP model developed by researchers at OpenAI. The model is the standard version of vision encoders providing effective representation of the image features. The clip-L is used in such multimodal models as LLaVA-v1.5, LLaVA-Next, InternLM-XComposer2-VL [28] and others.
3. SigLIP/16-512[2]. The SigLIP model was proposed in [8] by Google Research. SigLIP proposes to replace the loss function used in CLIP with a simple pairwise sigmoid loss. The model outperforms most of the existing models so far and provides state-of-the-art results in the classification tasks.
4. InternViT-6B[3]. InternViT-6B-448px-V1-2 [29] is the largest open-sourced vision/vision-language foundation model to date, achieving 32 state-of-the-art performances on a wide range of tasks such as visual perception, cross-modal retrieval, multimodal dialogue, etc. The model was proposed by the OpenGVLab team.

   Pretrain Dataset: LAION-en, LAION-COCO, COYO, CC12M, CC3M, SBU, Wukong, LAION-multi, OCR data.

   The VLMs that utilize such encoders are as follows: InternVL-Chat-V1.2, InternVL-Chat-V1.2-Plus.

Table 3 provides the overall description of the observed vision encoders that can be used for vision encoding in multimodal models.

Table 3: Vision encoders characteristics.

	patch resolution	fine-tuning resolution	num hidden layers	image tokens num	emb size	emb layer
CLIP ViT-bigG/14	14x14	224x224	48	256	1664	-2
CLIP VIT-large/14	14x14	336x336	24	576	1024	-2
SigLIP-base/16-512	16x16	512x512	12	1024	768	-2
InternViT-6B-448px-V1-2	14x14	448x448	45	1024	3200	-1

To evaluate the performance of different visual encoders on the multimodal model's visual understanding abilities, we trained the final model using the same data samples but with various vision backbones. We then assessed the model's quality on multimodal benchmarks to compare the effectiveness of different visual encoders. We present the evaluation results of OmniFusion trained with different vision backbones in Table 4.

Table 4: Comparison of OmniFusion performance with different vision backbones on multimodal benchmarks.

	Resolution	ScienceQA-full	ScienceQA-img	text-VQA	VQAv2	VizWiz	POPE	MMBench	MMMU	MM-Vet
CLIP ViT-bigG/14	336	68.71	66.68	47.58	73.28	49.92	82.80	61.73	30.60	30.10
CLIP VIT-large/14	336	71.21	68.17	57.20	78.42	**54.11**	84.50	65.72	30.90	33.60
SigLIP-base/16-512	512	71.92	68.62	57.05	78.88	51.84	**86.5**	66.80	31.60	33.70
InternViT-6B-448px-V1-2	448	**73.38**	**69.71**	**61.57**	**80.08**	52.78	84.47	**67.10**	**35.40**	**35.10**

Notably, the best performance is achieved by OmniFusion trained with the largest visual encoder, InternViT-6B-448px-V1-2. Despite the higher resolution for SigLIP, CLIP VIT-large/14 and SigLIP-base/16-512 achieve comparable results on most tasks. However, both of them significantly outperform CLIP ViT-bigG/14 with the smallest resolution.

Mix of image encoders. Rather than relying solely on a single image encoder, one approach to enhance image representation is to merge embeddings from multiple encoders. In this section, we considered various image encoders and adapters for mixing encoder features. We trained a language model and adapter using the features from multiple image encoders simultaneously. OpenAI CLIP-ViT-L/14 (336×336) and ViT-DINO-v2 models were used as image encoders. Mixing of encoder features was performed in the adapter module. The following 4 different adapter module architectures were investigated.

The first architecture (baseline) corresponds to the encoder and adapter used in the previous experiments. It uses only one encoder (OpenAI CLIP-ViT-L/14) and an adapter with a transformer decoder layer. More specifically, the adapter unit has the following architecture. The model first uses the transformer encoder layer and then does a linear transformation to change the dimension of the embedding vectors.

The second architecture (embedding concatenation) uses two encoders, first, the vectors of the feature sequences from the penultimate layers of encoders are independently linearly mapped. Two different linear transformations are used for the features from the first and the second encoders. Afterward, the embedding sequences are then concatenated and fed to the encoder layer of the transformer and mapped linearly (to change the dimension of the embeddings) as in the baseline.

[2]https://github.com/merveenoyan/siglip

[3]https://github.com/OpenGVLab/InternVL-MMDetSeg

The third architecture, inspired by COMM [30], also uses 2 encoders. An important difference from the previous version is the use of the features from all layers of image encoders. First, layer normalization is applied to the feature from all layers of each encoder, then linear transformation is applied (different mappings are used for the features from the different layers), after which, a linear combination of the features from different layers is calculated (the coefficients of the linear combination are trainable parameters), the resulting features for two encoders are summed (sequences of embeddings from two encoders have different lengths, so the shorter sequence is concatenated with zero vectors before the addition). Finally, the GELU activation function and the final linear transformation are applied.

The last architecture, the fourth one, is similar to the third, but after aggregating features from different layers of encoders (counting linear combinations of features from different layers of encoders), it uses a different method of mixing features from two encoders. GELU is applied to the embedding sequences from two encoders, then the embedding sequences are concatenated and fed to the multi-head attention layer as a query, and a trained parameter matrix is used for the key and value. In the end, the GELU transform and linear mapping are applied. This approach is interesting because by changing the size of the trained parameter matrix, it is possible to change the number of vectors in the output embedding sequence. We used a matrix of 576 rows so that the length of the visual embedding sequence is the same as in the baseline.

Table 5 presents the results of the tests on various benchmark models that use different methods of mixing features from two visual encoders.

Table 5: Comparison of OmniFusion performance with different CLIP VIT-L/14 and DinoV2 encoders mix.

	ScienceQA-full	ScienceQA-img	Text-VQA	VQAv2	VizWiz	POPE	MMBench	MMMU	MM-Vet
Baseline (one encoder)	71.21	68.17	57.20	78.42	54.11	84.50	65.72	32.30	33.60
Embeddings concatenation	**72.25**	**68.62**	54.02	75.95	51.33	85.40	67.87	34.10	30.60
Projection & summation	71.09	67.18	**60.87**	**78.77**	**57.09**	**85.73**	**69.24**	**36.50**	**34.60**
Projection & multi-head attention	72.01	68.17	47.75	70.84	50.82	84.93	57.47	29.40	21.90

The best method turned out to be one that uses features from all the layers of encoders and sums them to mix the features from two encoders. We also noticed that the last approach with a multi-headed attention layer turned out to be the worst in terms of the inference speed.

In the initial experiments, we also compared two adapter techniques: linear projection, implemented as a two-layer MLP with GeLU activation, and a simple transformer-based Encoder block with one layer and four heads. We found that the best results were obtained with the simple linear projection. While the transformer-based adapter performs comparably to MLP on some benchmarks (such as VizWiz or ScienceQA), we observed a drop of up to 4 points on OCR-based tasks.

We hypothesize that this difference can be explained by the fact that originally, Vision Transformer (VIT) models trained in the CLIP pipeline are naturally aligned with text, making linear transformation suitable for this task. Additionally, we speculate that larger trainable adapters require more data for successful alignment.

Scaling Images to HD. Following the methodology of LLaVA-NeXT [9], we investigate whether image resolution plays a significant role in capturing fine-grained pixel scale information.

Our approach involved resizing images adaptively to the input size and dividing them into non-overlapping sections based on the resolution of the image encoder. This strategy ensured that we retained all the specific details from the original images, thus enhancing the model's performance on tasks that rely on such details.

To encode the image, we employ CLIP-ViT-L/14 (336×336), treating each patch as a separate entity. Initially, we resized and padded the image to the most suitable dimensions. Then, we divided it into a grid of 336×336 patches, encoding each patch individually. As has been observed in previous research, encoding the entire image into a sequence of image tokens is crucial for preserving the overall context of the scene, so, we also kept this image representation in our final experiment.

To assess the model's capability to encode images with higher resolution. we trained three versions of the same model. The baseline model is OmniFusion-7B with the frozen OpenAI Clip-L vision encoder. The second model follows the same architecture, but during fine-tuning, the image is split into patches and encoded by the Clip-L encoder. Finally, we augmented the training data with 20K proprietary samples from the DocVQA domain to evaluate whether this additional dataset, combined with image splitting, enhances the model's performance in the document domain.

Table 6 presents the results of OmniFusion trained with the strongest vision backbone in standard resolution (448×448), and in high resolution with CLIP-ViT-L/14 (336×336), and grid split on standard multimodal benchmarks, while Table 7 shows the results for specific benchmarks related to document and infographics analysis.

Table 6: Comparison of OmniFusion performance w and w/o grid splitting of the input image.

	ScienceQA full	ScienceQA img	text-VQA w OCR	Text-VQA w/o OCR	VQAv2	POPE	MMBench	MMMU	MM-Vet	GQA
OmniFusion InternViT-6B-448px-V1-2	**73.38**	**69.71**	61.57	56.48	80.08	84.47	67.10	35.40	35.10	63.44
OmniFusion grid split	71.23	67.97	63.31	60.06	80.86	86.72	**69.00**	34.10	36.60	64.18
OmniFusion grid split + ruDocVQA	73.21	69.16	**65.53**	**63.50**	**80.94**	**87.21**	67.61	**36.60**	**39.40**	**64.57**

Table 7: Comparison of OmniFusion w and w/o grid splitting for document and infographics analysis.

	InfoVQA	ChartQA	DocVQA	multiDocVQA
OmniFusion InternViT-6B-448px-V1-2	29.24	19.92	38.44	22.5/11.43
OmniFusion grid split	30.2	20.2	46.76	27.09/14.5
OmniFusion grid split + rudocVQA	**36.99**	**28.96**	**62.82**	**36.38/20.55**

Based on the observed results, we make the following observations:

1. First, high image resolution is particularly relevant in domains such as OCR. Additionally, the evaluation of hallucination (POPE) is also influenced by image resolution.
2. Another interesting finding concerns document-based benchmarks. Adding 20K samples of proprietary document-level data in Russian significantly boosted the model's performance on document analysis (refer to Table 7, line 3).

Tuning with synthesized TeX formulas. In this section, we examined SFT for the task of generating LaTeX code from a formula image. The datasets used for training included CROHME [31] (containing pairs of handwritten formulas and LaTeX code) and the im2latex dataset (containing pairs of formulas rendered in different fonts and their corresponding LaTeX code). Both of these preprocessed datasets were obtained from the open-source repository LaTeX-OCR[4]. Overall, the character sequence length in the training dataset ranges from 1 to 4296 symbols.

As image encoders we used CLIP VIT-large/14 and Donut [32] vision encoder. The last was fine-tuned for the text and LaTeX formula recognition in the open-source repository texify[5] and provided a good latent representation for this specific case of math equation recognition.

SFT stage consisted of two steps:

1. Freezing the image encoder and LLM, fine-tuning the adapter and the special tokens.
2. Freezing the image encoder, fine-tuning the adapter, and the special tokens, and fine-tuning LLM using LoRA.

As a result of the study, when using CLIP VIT-large/14 as the image encoder, the model performs poorly in long formula recognition but is capable of perceiving the handwritten ones, whereas the Texify vision encoder shows the opposite behavior. The comparison of the two encoders for TeX formula recognition is presented in Table 8.

The normalized edit distance (NED) was utilized as a metric to evaluate the quality of LaTeX code generation across 30,600 samples from the im2latex test dataset, encompassing formulas ranging in length from 4 to 2226 symbols.

Table 8: Comparison of two image encoders for OmniFusion for TeX formulas recognition.

Image Encoders	NED ↓
CLIP VIT-large/14	0.32
Texify vision encoder	**0.19**

Figure 4 provides examples of LaTeX formula understanding by OmniFusion with the Texify vision encoder.

[4]https://github.com/lukas-blecher/LaTeX-OCR
[5]https://github.com/VikParuchuri/texify

$$\boxed{\sum_{j=1}^{n}|\mathbf{a_j}|^2=\sum_{j=1}^{n}\sum_{i=1}^{r}(\mathbf{a_j}\cdot\mathbf{v_i})^2=\sum_{i=1}^{r}\sum_{j=1}^{n}(\mathbf{a_j}\cdot\mathbf{v_i})^2=\sum_{i=1}^{r}|A\mathbf{v_i}|^2=\sum_{i=1}^{r}\sigma_i^2(A).}$$

$\downarrow$

\sum _ { j = 1 } ^ { n } \left| \mathbf { a } _ { j } \right| ^ { 2 } = \sum _{j = 1} ^{ n } \sum _{i = 1}^{n } \left(\mathbf {a } _ { \mathbf {j } } \cdot \mathbf {v } _ { 1 } \right) ^ {2} = \sum_{i=1}^ {n} \sum_{j=1}^{n} (\mathbf{a } _ {\mathbf{j} } \cdot { \mathbf{v } } _ {1 }) ^ {n} = \left(i = 1 \right) = \left| A \mathbf{ v } _ {{ 1 }} \right| ^{2} = { \sum_{ i =1 } ^{n} } \sigma _ { i } ^{2} (A) .

$\downarrow$

$$\textstyle\sum_{j=1}^{n}|\mathbf{a}_j|^2=\sum_{j=1}^{n}\sum_{i=1}^{n}(\mathbf{a_j}\cdot\mathbf{v}_1)^2=\sum_{i=1}^{n}\sum_{j=1}^{n}(\mathbf{a_j}\cdot\mathbf{v}_1)^n=(i=1)=|A\mathbf{v}_1|^2=\sum_{i=1}^{n}\sigma_i^2(A).$$

$$\boxed{q^\star\triangleq\arg\min_q\mathbb{E}\Big[\big\|q(z_\theta)-z'_\xi\big\|_2^2\Big],\quad\text{where}\quad q^\star(z_\theta)=\mathbb{E}\big[z'_\xi|z_\theta\big],}$$

$\downarrow$

q ^ { \star } \triangleq \operatorname { a r g } \operatorname * { m i n } _ { q } \mathbb { E } \Bigl [\bigl \| q (z _ { \theta }) - z _ {\xi } ^ { \prime } \bigr \| ^ { 2 } \Bigr] , \quad \mathrm { w h e r e } \quad q ^ { * } (z _ {\theta }) = \operatorname { E } [z _ {\zeta } | z _ {\vartheta }] ,

$\downarrow$

$$q^\star\triangleq\arg\min_q\mathbb{E}\Big[\big\|q(z_\theta)-z'_\xi\big\|^2\Big],\quad\text{where}\quad q^*(z_\theta)=\mathrm{E}[z_\zeta|z_\vartheta],$$

Figure 4: An example of LaTeX formula understanding by OmniFusion fine-tuned with the Texify vision encoder is depicted below. The upper image displays the input image, while the lower image showcases the compilation of LaTeX code generated by the model.

3.2 Evaluation on benchmarks

In this section, we present the evaluation results of the introduced model. Table 9 presents a comparison of existing leading approaches to training multimodal models with our best options of OmniFusion. We provide a comparison of the models with standard and high resolution across several benchmarks in zero-shot settings. For the evaluation of OmniFusion and LLaVA, we utilize the recent open-source library lmms-eval [33].

Main results. The main results are as follows: at standard resolution, OmniFusion using the largest vision encoder InternViT-6B-448px-V1-2 achieves the best scores on most of the benchmarks, being on an equal basis with models based on larger LLMs (such as Vicuna-13B [37]).

Moreover, in the subgroup of models that operate on images with the same resolution, the proposed approach of encoder mixing significantly improves results in Text-VQA and GQA datasets, adding one point, on average, to the MMMU benchmark as well.

For high resolution, the version of OmniFusion with grid splitting provides comparative results with state-of-the-art open-source multimodal models.

4 Related Work

Taking advantage of LMMs powerful capability to follow instructions, active work is underway to extend their ability to perceive other types of data, particularly visual data. Early work in the field, including FROMAGe [38], LLaVA [39], MiniGPT4 [40], and BLIP-2 [41], encoded visual information into one or more visual tokens by training only a lightweight projection that efficiently matched images to the language model. This approach has enabled tasks such as image captioning and visual question answering to be partly solved. Several subsequent studies have focused on enhancing and adjusting the projection architecture [42, 1, 43, 44], and determining the need for pretraining LLM on image text data [45, 2, 34, 46]. Another promising approach was introduced in [47, 48], the authors showed that being pre-trained from scratch on large amount of interleaved data multimodal models obtain external abilities to understand

Table 9: Comparison with recent multimodal models on zero-shot benchmarks. The best results across model subgroups are in bold; the best performance for models with LLMs of the same size is underlined. As we lack information on how TextVQA is assessed, for some models we include additional evaluation with supported OCR tokens in brackets. We do not provide information on whether the train set of the benchmark was included.

Model	LLM	Encoder	Res.	ScienceQA full	ScienceQA img	Text-VQA	VQAv2	POPE	GQA	MMMU	MM-Vet
Standard resolution											
LLaVA-v1.5 13B [18]	Vicuna-13B	CLIP VIT-large/14	336	**74.96**	**72.88**	48.73 (60.75)	79.52	85.92	63.24	34.80	36.70
LLaVA-v1.5 7B [18]	Vicuna-7B	CLIP VIT-large/14	336	70.41	70.43	46.07 (58.20)	78.50	85.87	61.97	35.30	30.55
Qwen-VL-Chat [34]	Qwen-7B	CLIP VIT-bigG/14	448	-	68.2	63.80	-	**88.10**	57.50	**37.00**	**47.30**
Mini-Gemini [35]	Vicuna-7B	CLIP VIT-large/14 ConvNeXt-L	336	-	-	**65.20**	-	**88.10**	-	36.10	40.80
OmniFusion	GigaChat-7B	CLIP VIT-large/14	336	71.21	68.17	51.87 (57.20)	78.42	84.50	**65.72**	32.30	33.60
OmniFusion	GigaChat-7B	InternViT-6B-448px V1-2	448	74.13	71.29	56.48 (61.57)	**80.08**	84.14	63.44	35.40	35.10
Encoder Merge											
OmniFusion	GigaChat-7B	CLIP VIT-large/14 +DinoV2	336	71.09	67.18	49.75 (60.87)	78.77	85.73	64.59	36.50	34.60
OmniFusion	Mistral-7B	CLIP VIT-large/14 + DinoV2	336	70.30	67.55	48.93 (59.15)	-	82.18	64.00	35.90	33.10
OmniFusion	Vicuna-7B	CLIP VIT-large/14 + DinoV2	336	71.26	69.96	49.58 (61.04)	-	85.81	62.60	36.90	33.60
HR resolution											
LLaVA-Next 7B	Vicuna-7B	CLIP VIT-large/14	672	73.21	70.15	64.85	80.06	86.40	64.23	35.10	**44.08**
LLaVA-Next 13B	Vicuna-13B	CLIP VIT-large/14	672	**75.85**	**73.57**	**66.92**	80.92	86.26	**65.36**	35.90	**49.12**
DeepSeek-VL [36]	DeepSeek-7B	SigLIP-L SAM-B	384 1024	-	-	-	-	88.10	-	36.6	41.50
Mini-Gemini-HD [35]	Vicuna-7B	CLIP VIT-large/14 ConvNeXt-L	672	-	-	68.40	-	-	-	**36.8**	41.30
OmniFusion + grid-split	GigaChat-7B	CLIP VIT-large/14	672	73.21	69.16	63.5 (65.53)	**80.94**	**87.21**	64.57	36.60	39.40
Proprietary models											
GPT-4V	Unk	Unk	Unk	-	-	78.00	-	-	-	**56.80**	49.90
Gemini Pro	Unk	Unk	Unk	-	-	74.60	-	-	-	47.90	**64.30**
Qwen-VL-Plus	Unk	Unk	Unk	-	-	78.90	-	-	-	45.20	55.70
Qwen-VL-Max	Unk	Unk	Unk	-	-	**79.50**	-	-	-	51.40	61.80

long context and support few-shot learning. However, since training these models requires a lot of computational resources there's greater interest in lighter-weight approaches. These involve partially freezing specific parts of the backbones within the multimodal pipeline [49, 39, 34].

Significant improvement in visual-textual perceptual quality was achieved by curating more synthetic fine-tuning data, as introduced in LLaVA-1.5 [18]. This model has a relatively simple architecture and uses only publicly available datasets, highlighting the importance of data and multimodal task diversity for VLM training. A number of subsequent papers have improved this approach by enhancing the visual grounding capability in LLaVA-G [50], adapting to specific domains as LLaVA-Med [5]; instructively learning to use tools and invoking other task-specific models in LLaVA-PLUS [4], improving efficiency through MoE-based sparse modification in MoE-LLaVA [51], and reducing the underlying language model to Phi-2.7B in the LLaVA-Phi paper [52]. The Video-LLaVA work [3] continues to expand the horizons of multimodality to the video comprehension.

At the same time, state-of-the-art results have been achieved through careful investigation of the most beneficial process of visual-textual pre-learning and the effective incorporation of the visual modality into VLLM. The VILA paper [53] argues that unfreezing the LLM allows for stronger in-context learning and multi-image reasoning capabilities. In addition, these results also further support the conclusions of the Monkey paper that the image resolution is more important than the number of visual tokens. Such an approach to embedding visual modality may help to provide greater scalability and better quality on fine-grained visual-language tasks. COMM [30] revised the effectiveness of already existing visual models in MLLM and proposed a simple yet effective multi-level features merging strategy for different image encoders to enhance visual capabilities. The follow-up VistaLLM work [54] proposed an instruction-guided image tokenizer that filters global embeddings using task descriptions to extract compressed and refined features from images into a unified general-purpose model.

Despite these advancements, there are still open challenges in extracting better visual features from images, indicating the need for further development of multimodal architectures in this area. Our work takes mutual advantages of both previously mentioned research directions, i.e., using the LLaVA-based VLM architecture as a baseline and effectively combining multiple visual encoders into a single visual branch providing a good opportunity to enhance latent visual representation for general and specific domains.

5 Conclusion

In conclusion, our OmniFusion model represents significant progress in the field of multimodal learning, integrating textual and visual data within a single framework. By evaluating various architecture design principles, and image encoding methods, and integrating multiple vision encoders, we have demonstrated the model's superior performance across a wide array of visual-language benchmarks. Our findings underscore the effectiveness of OmniFusion in handling complex VQA tasks, outperforming existing solutions, and providing detailed, domain-specific responses. With the release of our open-source Mistral-based OmniFusion model, including weights and scripts for training and inference, we aim to contribute to the broader AI research community, facilitating further developments in multimodal AI systems.

Further development of our research will shed light on the new approaches of efficient image embedding extraction including not only encoders research but also some hierarchical image representation methods. In terms of overall architecture design, we plan to go deeper into long image context processing, especially in video modality integration. At the same time, providing fast and reliable image output generation using our latent diffusion model Kandinsky [55] is also one of the major research tasks, which will lead to the new OmniFusion features like local image editing by text prompt, image generation during multimodal dialog context, etc.

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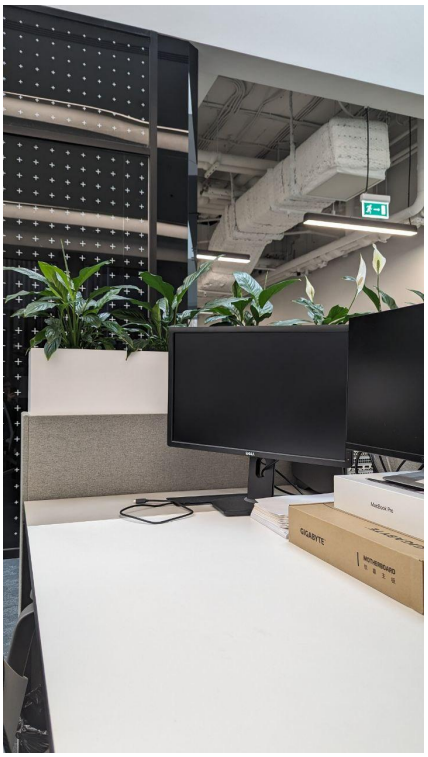
B. OmniFusion Examples

Below there are few examples of OmniFusion generations in English and Russian.

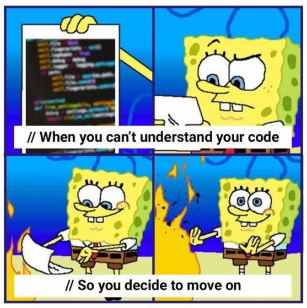


What kind of plant is shown in the background?

In the background of the image, there is a plant with large green leaves and white flowers. This plant is most likely a peace lily (Spathiphyllum wallisii), which is a popular indoor plant known for its ability to thrive in low-light environments.

What is the algorithm for caring for him?

The algorithm for caring for a peace lily is as follows: 1. Place the plant in an area with bright, indirect light. 2. Water the plant when the top inch of soil feels dry to the touch. 3. Fertilize the plant once a month during the growing season (spring and summer) with a balanced fertilizer. 4. Prune the plant as needed to maintain its desired shape and size. 5. Keep the plant away from cold drafts and direct sunlight, as these can damage the leaves. 6. Clean the leaves regularly to remove dust and debris, which can block the plant's ability to photosynthesize. 7. Repot the plant every 2-3 years as it grows, using a well-draining potting mix and a pot that is slightly larger than the current one.



Explain this meme.

The meme features a cartoon image of SpongeBob SquarePants holding a piece of paper with code written on it. The text above the image reads, "When you can't understand your code," and the text below the image says, "So you decide to move on." This meme is a humorous way of expressing the frustration and difficulty that many programmers face when trying to understand and debug their code. The decision to "move on" is often a necessary step in the process of problem-solving, as it allows the programmer to take a step back and approach the problem from a different angle or with a fresh perspective.

Tell me the breed of this dog

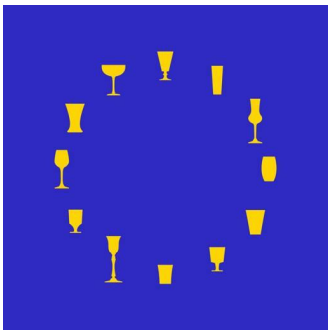
The dog in the image is a beagle.

How should one take care of this dog?

To take care of a beagle, it is important to provide them with regular exercise, mental stimulation, and a healthy diet. Beagles are energetic and playful dogs, so they need plenty of opportunities to run, play, and explore. Regular walks, trips to the park, and games of fetch can help satisfy their need for physical activity. Additionally, beagles are intelligent and curious dogs, so providing them with interactive toys, puzzles, and training sessions can help keep them mentally stimulated and prevent boredom. Lastly, a healthy diet is essential for a beagle's overall well-being. Beagles should be fed high-quality dog food that is appropriate for their age, size, and activity level. It is also important to avoid overfeeding, as beagles are prone to weight gain and obesity.

What is the origin of this breed?

The beagle is a breed of dog that originated in England. They were originally bred for hunting small game, such as rabbits and hares, and were known for their keen sense of smell and ability to follow a scent trail. Over time, the beagle has become a popular companion dog, known for their friendly and affectionate nature.



Describe what you see in the image in detail

The image presents a circle made up of 12 wine glasses, arranged in a circular pattern. The glasses are of varying sizes and shapes, creating a visually interesting composition. The background of the image is a solid blue color, providing a stark contrast to the yellow wine glasses. The arrangement of the glasses suggests a sense of unity and harmony, as if they are all part of a larger whole.

What should I cook from this for family dinner?

Based on the image, you could cook a variety of dishes for your family dinner using the items in the refrigerator. Some suggestions include: 1. Roast turkey: You have a whole turkey in the fridge, so you could roast it for a traditional Thanksgiving or holiday meal. 2. Salad: The lettuce, tomatoes, cucumbers, and peppers can be used to make a fresh and healthy salad. 3. Stir-fry: You can use the chicken, peppers, and onions to make a delicious stir-fry dish. 4. Soup: The chicken, vegetables, and broth can be combined to make a hearty and nutritious soup. 5. Sandwiches: You could use the turkey, cheese, and bread to make sandwiches for a quick and easy meal. These are just a few ideas, and you can mix and match the ingredients to create your own unique dishes. It's important to consider your family's preferences and dietary restrictions when planning the meal.

Sounds nice, but I don't have cheese

Sounds nice, but I don't have cheese
If you don't have cheese, you can still make a variety of dishes using the items in the refrigerator. Some suggestions include: 1. Roast turkey: You have a whole turkey in the fridge, so you could roast it for a traditional Thanksgiving or holiday meal. 2. Salad: The lettuce, tomatoes, cucumbers, and peppers can be used to make a fresh and healthy salad. 3. Stir-fry: You can use the chicken, peppers, and onions to make a delicious stir-fry dish. 4. Soup: The chicken, vegetables, and broth can be combined to make a hearty and nutritious soup. 5. Sandwiches: You could use the turkey and bread to make sandwiches for a quick and easy meal. These are just a few ideas, and you can mix and match the ingredients to create your own unique dishes.

Как ты думаешь, к чему приурочен этот постер?

Я думаю, что этот постер приурочен к автомобильной гонке в Японии.

Какого числа состоится гонка?

5 апреля 2021 года.

Болид чьей команды изображен на постере?

Я думаю, что на постере изображена команда Ferrari.

Молодец!

Спасибо!

Что можно приготовить из этих продуктов?

Что можно приготовить из этих продуктов?
На картинке представлены различные продукты питания, включая яйца, муку, сахар, масло, молоко, соль и другие. Их можно использовать для приготовления разнообразных блюд, таких как омлеты, блины, пироги, торты, печенье, соусы и многое другое.

Напиши рецепт блинов.

Рецепт блинов

Ингредиенты:
1 стакан муки;
1 стакан молока;
1 яйца;
1 ст. л. сахара;
1/2 ч. л. соды или разрыхлителя;
2 ст. л. подсолнечного масла.

Приготовление:

1. В миске смешайте муку, сахар и соду.
2. Добавьте яйцо и молоко, хорошо перемешайте.
3. Добавьте масло и снова перемешайте.
4. Разогрейте сковороду на среднем огне и смажьте её маслом.
5. Налейте небольшое количество теста на сковороду и равномерно распределите его по всей поверхности сковороды.
6. Жарьте блины до золотистого цвета с обеих сторон.
7. Готовые блины выложите на тарелку и подавайте на стол.

Приятного аппетита!

Из какой игры эти герои?

Эти герои являются персонажами из игры Minecraft.

Который час?

На картинке изображены часы, на которых отчетливо видны цифры 21:42.

Посмотри на изображение и ответь в следующей форме: 1. Название предмета 2. Алгоритм его сборки 3. Примерное время сборки в часах

1. Название предмета: велосипед.
2. Алгоритм его сборки: сначала нужно собрать раму, затем установить переднее и заднее колеса, после этого следует установить руль и тормоза.
3. Примерное время сборки в часах: 2-3 часа.